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Chapter 1

Introduction to the problem

1.1 Problem 25 ML

25. (P) is so-called *extreme learning*, i.e., a neural network with one hidden layer, $y = w\sigma(W_1x)$, where the weight matrix for the hidden layer W_1 is a fixed random matrix, $\sigma(\cdot)$ is an elementwise activation function of your choice, and the output weight vector w is chosen by solving a least-squares problem with L_1 regularization $\arg \min_w f(w)$, with $f(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|w\|_1$.

(A1) is *iteratively reweighted least squares*: i.e., an iteration where you solve at each step the least-squares problem

$$w_{k+1} = \arg \min_w f_k(w), \quad f_k(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|W_k w\|_2^2$$

where W_k is the diagonal matrix with entries $(W_k)_{ii} = |(w_k)_i|^{-1/2}$.

(A2) is an algorithm of the class of the *deflected subgradient methods*.

1.2 Extreme Learning Machine

An Extreme Learning Machine [18] is a single-hidden-layer feedforward network in which the input-to-hidden weights are fixed at random and only the hidden-to-output weights are trained. We fix the notation: $\mathbf{W}_1 \in \mathbb{R}^{H \times N}$ is the random, frozen input-to-hidden matrix (H hidden units, N inputs); $\sigma(\cdot)$ is an elementwise nonlinearity, taken to be $\tanh$; $\mathbf{w} \in \mathbb{R}^H$ is the trainable output vector; the model output is $\hat{y} = \mathbf{w}^T \sigma(\mathbf{W}_1 \mathbf{x}) \in \mathbb{R}$. Since $\mathbf{W}_1$ is fixed, training $\mathbf{w}$ reduces to a (regularized) linear least-squares problem in the hidden-layer features.

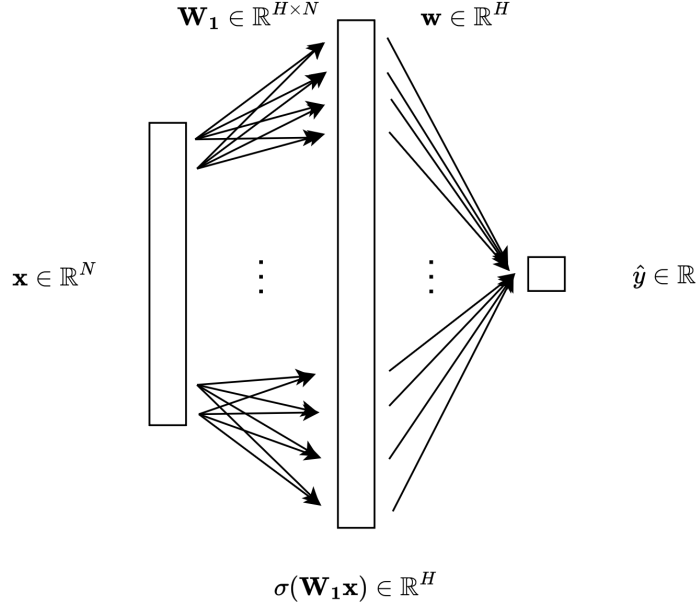


Figure 1.1: Structure of the ELM model used in this project.

1.3 Least-Squares Problem

Let $\mathbf{X}_{\text{origin}} \in \mathbb{R}^{M \times N}$ be the matrix of M training inputs stacked as rows, $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T) \in \mathbb{R}^{M \times H}$ the hidden-layer activations, and $\mathbf{y} \in \mathbb{R}^M$ the target vector. Training $\mathbf{w}$ amounts to solving [3]

$$\min_{\mathbf{w}} \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2, \quad (1.1)$$

which under the full-column-rank assumption on $\mathbf{X}$ admits the unique closed-form solution $\mathbf{w}_0 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. Adding the L_1 regularization term (Section 1.4) breaks this closed form and forces iterative methods.

1.4 Regularization Techniques

1.4.1 L_1 regularization

The project statement prescribes an L_1 penalty on $\mathbf{w}$ [11], giving the regularized objective

$$\min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \sum_{i=1}^H |w_i|. \quad (1.2)$$

Here λ_{LASSO} denotes the regularization parameter called λ in the project statement (Section 1.1); a related symbol $\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}}$, induced by the IRLS

majorization, will be introduced in Chapter 2. The L_1 term makes f non-differentiable wherever any $w_i = 0$, ruling out plain gradient methods and motivating the two algorithms developed in the next chapters (IRLS, Chapter 2; deflected subgradient, Chapter 3).

1.5 Objective Function Analysis

The full objective function we need to minimize is the Lasso regularization of the Least-Squares problem:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})}$$

It is useful to analyse the mathematical properties of $f(\mathbf{w})$, as they determine which algorithms can be applied.

1.5.1 Convexity

f is the sum of two convex functions [6]. The quadratic term $g(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$ has Hessian $\mathbf{X}^T \mathbf{X} \succeq 0$, so it is convex; under the ELM full-column-rank assumption the Hessian is positive definite and g is in fact *strictly* convex. The ℓ_1 penalty $h(\mathbf{w}) = \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$ is convex as a scaled norm, but it is *not* strictly convex (the segment between two sign-aligned vectors keeps it affine). Strict convexity of $f = g + h$ therefore relies entirely on g : whenever g is strictly convex, so is f , and the minimizer $\mathbf{w}^*$ is unique.

Proposition 1.1. *If $\mathbf{X} \in \mathbb{R}^{M \times H}$ has full column rank, then $f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$ is strictly convex and admits a unique global minimizer $\mathbf{w}^*$.*

1.5.2 Non-smoothness

Unlike the quadratic term, the L_1 penalty $h(\mathbf{w}) = \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$ is **not differentiable** [13] at any point where some component $w_i = 0$. This means f itself is non-smooth, and we cannot rely on gradient-based methods alone. At points where all $w_i \neq 0$, the gradient exists and is:

$$\nabla f(\mathbf{w}) = \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w})$$

where $\text{sign}(\mathbf{w})$ is applied element-wise. At points where some $w_i = 0$ one must reason in terms of *subgradients*, which we recall formally in the next subsection and use extensively in Chapter 3.

1.5.3 Optimality conditions

Since f is non-smooth, the classical condition $\nabla f(\mathbf{w}^*) = 0$ does not apply at every minimizer. Instead, the optimality condition is expressed via the **sub-differential** [9]: $\mathbf{w}^*$ is a global minimizer for our ELM problem if and only

if:

$$0 \in \partial f(\mathbf{w}^*) = \mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \partial \|\mathbf{w}^*\|_1$$

where the subdifferential of the L_1 norm is given component-wise by:

$$[\partial \|\mathbf{w}\|_1]_i = \begin{cases} \{+1\} & \text{if } w_i > 0, \\ \{-1\} & \text{if } w_i < 0, \\ [-1, +1] & \text{if } w_i = 0. \end{cases}$$

This condition can be written component-wise as: for each $i = 1, \dots, H$,

$$[\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y})]_i + \lambda_{\text{LASSO}} s_i = 0, \quad s_i \in \partial |w_i^*|. \quad (1.3)$$

Intuitively, λ_{LASSO} controls the sparsity of $\mathbf{w}^*$: the larger λ_{LASSO} is, the easier it is for the optimality condition to force components $w_i^* = 0$, producing a sparser solution.

Chapter 2

Algorithm 1: Iteratively Reweighted Least Squares

Iteratively Reweighted Least Squares (IRLS) [4] is a classical algorithm for solving optimization problems involving non-smooth objective functions of the form of a p -norm. Its core idea is to handle non-smoothness by solving a sequence of *weighted* least-squares problems, each of which is smooth and admits an efficient closed-form solution. In our setting, IRLS is used to minimize the L_1 -regularized least-squares objective:

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})} \quad (2.1)$$

where $\mathbf{X} \in \mathbb{R}^{M \times H}$, $\mathbf{y} \in \mathbb{R}^M$, and $\lambda_{\text{LASSO}} > 0$ is the regularization hyperparameter of the original problem.

2.1 The core idea: a strict upper bound for the L_1 norm

The difficulty in (2.1) is the L_1 term: convex but non-differentiable at every component crossing zero. To replace it with a smooth surrogate we use the elementary inequality $(|w_i| - |(w_k)_i|)^2 \geq 0$, which after expansion and division by $2|(w_k)_i|$ gives the majorization

$$|w_i| \leq \frac{w_i^2}{2|(w_k)_i|} + \frac{|(w_k)_i|}{2}, \quad (w_k)_i \neq 0. \quad (2.2)$$

Summing over $i = 1, \dots, H$ yields the bound on the L_1 norm

$$\|\mathbf{w}\|_1 \leq \frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w} + \frac{1}{2} \|\mathbf{w}_k\|_1 \quad (2.3)$$

where $\mathbf{W}_k \in \mathbb{R}^{H \times H}$ is a diagonal matrix with entries:

$$(\mathbf{W}_k)_{ii} = |(w_k)_i|^{-1/2} \quad (2.4)$$

This bounds the non-smooth L_1 term using a *smooth, quadratic* expression $\frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$, plus a constant term that depends only on the current iterate $\mathbf{w}_k$. Intuitively, $\mathbf{W}_k^T \mathbf{W}_k$ implements a *personalized penalization* of each component of $\mathbf{w}$ [2]: components with small magnitude at iteration k receive a large weight $|(w_k)_i|^{-1}$ and are strongly pushed toward zero, while components with large magnitude receive a small weight and are left relatively free. This *reweighting* mechanism is what drives IRLS to produce sparse iterates, mimicking the sparsity-inducing effect of the L_1 norm through a sequence of smooth quadratic problems.

Handling near-zero components. Equation (2.4) becomes numerically unstable when $(w_k)_i \approx 0$, since $|(w_k)_i|^{-1/2}$ can blow up. To prevent division by (near) zero, a **safety threshold** $\varepsilon_{\text{thr}} > 0$ is introduced:

$$(\mathbf{W}_k)_{ii} = \left(\max(|(w_k)_i|, \varepsilon_{\text{thr}}) \right)^{-1/2} \quad (2.5)$$

Typical values are $\varepsilon_{\text{thr}} \in [10^{-6}, 10^{-10}]$.

2.2 The Majorization-Minimization interpretation

IRLS fits naturally inside the Majorization-Minimization (MM) framework [19]. Instead of minimizing f directly, at each iteration k we build a *surrogate* $Q(\mathbf{w}, \mathbf{w}_k)$ that (i) majorizes f , i.e. $Q(\mathbf{w}, \mathbf{w}_k) \geq f(\mathbf{w})$ for all $\mathbf{w}$; (ii) touches f at $\mathbf{w}_k$, i.e. $Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$; and (iii) is smooth, so that it can be minimized in closed form. Plugging the quadratic upper bound (2.3) into (2.1) gives our surrogate,

$$Q(\mathbf{w}, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1, \quad (2.6)$$

which is indeed a quadratic in $\mathbf{w}$. The majorization property is built in via (2.2). The touching property follows from a short computation: at $\mathbf{w} = \mathbf{w}_k$, each summand of the quadratic penalty becomes $(\mathbf{W}_k)_{ii}^2 (w_k)_i^2 = (w_k)_i^2 / |(w_k)_i| = |(w_k)_i|$, so

$$Q(\mathbf{w}_k, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w}_k - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 = f(\mathbf{w}_k).$$

Taking $\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} Q(\mathbf{w}, \mathbf{w}_k)$, the two MM properties combined give $f(\mathbf{w}_{k+1}) \leq Q(\mathbf{w}_{k+1}, \mathbf{w}_k) \leq Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$, so the sequence $\{f(\mathbf{w}_k)\}$ is non-increasing. IRLS thus guarantees a monotone decrease of the objective in exact arithmetic; in floating-point arithmetic the monotonicity is preserved up to the accuracy of the linear solver used for the subproblem.

2.3 The weighted least-squares subproblem

To find the next iterate $\mathbf{w}_{k+1}$, we must minimize the surrogate function $Q(\mathbf{w}, \mathbf{w}_k)$ with respect to $\mathbf{w}$:

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 \right)$$

Since the term $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1$ is constant with respect to $\mathbf{w}$, it does not affect the optimization and can be dropped entirely. The minimization problem simplifies to a standard **weighted least-squares problem with L_2 regularization** (Ridge regression):

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2 \right) \quad (2.7)$$

provided that we strictly define the IRLS regularization parameter as:

$$\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}} \quad (2.8)$$

Relation to the LASSO regularization parameter.

The factor $1/2$ in (2.8) is forced by the majorization constant in (2.3). The upper bound on $\|\mathbf{w}\|_1$ carries a coefficient $\frac{1}{2}$ in front of the quadratic form $\mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$, so the penalty entering the surrogate (2.6) is $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2$. Rewriting this as $\lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2$ in (2.7) therefore requires $\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}}$. This is precisely the factor that makes the fixed point of IRLS coincide with the LASSO optimum, as shown in the proof of Theorem 2.2.

Proposition 2.1 (Weighted normal equations). *The unique solution of (2.7) satisfies:*

$$(\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k) \mathbf{w}_{k+1} = \mathbf{X}^T \mathbf{y} \quad (2.9)$$

Proof. The gradient of the objective in (2.7) is:

$$\nabla f_k(\mathbf{w}) = \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$$

Setting this to zero yields (2.9). The coefficient matrix $\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ is symmetric positive definite (it is the sum of $\mathbf{X}^T \mathbf{X} \succeq 0$ and $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \succ 0$ since $\varepsilon_{\text{thr}} > 0$), so the system has a unique solution. $\square$

Since $\mathbf{W}_k$ is diagonal, defining $\mathbf{Q}_k = \mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ and $\mathbf{b} = \mathbf{X}^T \mathbf{y}$, equation (2.9) reduces to:

$$\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b} \quad (2.10)$$

This is an $H \times H$ symmetric positive definite (SPD) linear system, which can be solved efficiently at each iteration. Note that $\mathbf{b} = \mathbf{X}^T \mathbf{y}$ can be pre-computed once before the iterative loop, since it does not depend on k . Similarly, $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ is pre-computed once, and only the diagonal $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ is updated at each iteration.

Solving the linear system. Because $\mathbf{Q}_k$ is SPD, two efficient methods can be used:

- **Cholesky factorization:** [20] computes $\mathbf{Q}_k = \mathbf{L}\mathbf{L}^T$ in $O(H^3/3)$ operations, then solves two triangular systems in $O(H^2)$. This is the direct method of choice for moderate-sized problems.
- **Conjugate Gradient (CG):** [20] an iterative method that converges in at most H steps in exact arithmetic, with convergence rate depending on the condition number $\kappa(\mathbf{Q}_k)$. Preferable for large-scale problems where H is too large for Cholesky.

2.4 The complete algorithm

Algorithm 1 Iteratively Reweighted Least Squares (IRLS)

Require: $\mathbf{X} \in \mathbb{R}^{M \times H}$, $\mathbf{y} \in \mathbb{R}^M$, $\lambda_{\text{LASSO}} > 0$, $\varepsilon_{\text{thr}} > 0$, $\varepsilon_{\text{stop}} > 0$, k_{max}

- 1: Pre-compute $\mathbf{A} \leftarrow \mathbf{X}^T \mathbf{X}$, $\mathbf{b} \leftarrow \mathbf{X}^T \mathbf{y}$
- 2: Define $\lambda_{\text{IRLS}} \leftarrow \frac{1}{2} \lambda_{\text{LASSO}}$
- 3: Initialize $\mathbf{w}_0 \leftarrow \mathbf{A}^{-1} \mathbf{b}$ (ordinary least-squares solution)
- 4: **for** $k = 0, 1, 2, \dots, k_{\text{max}}$ **do**
- 5: Compute weights: $(\mathbf{W}_k)_{ii} \leftarrow (\max(|(w_k)_i|, \varepsilon_{\text{thr}}))^{-1/2}$ for all i
- 6: Assemble system: $\mathbf{Q}_k \leftarrow \mathbf{A} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$
- 7: Solve: $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$ (via *Cholesky* or *CG*)
- 8: **if** $\|\mathbf{w}_{k+1} - \mathbf{w}_k\| / \max(1, \|\mathbf{w}_k\|) < \varepsilon_{\text{stop}}$ **then**
- 9: **break** (convergence reached)
- 10: **end if**
- 11: **end for**
- 12: **return** $\mathbf{w}_{k+1}$

Initialization. We initialize $\mathbf{w}_0$ with the unregularized least-squares solution $\mathbf{A}^{-1} \mathbf{b}$: it is already available (we compute $\mathbf{A}$ and $\mathbf{b}$ once at the start anyway), the relative magnitudes of its components are a reasonable proxy for which weights the L_1 penalty will shrink, and no component is exactly zero, so the weights $(\mathbf{W}_0)_{ii} = |(w_0)_i|^{-1/2}$ are well defined without having to lean on ε_{thr} at the first step.

Stopping criterion. The algorithm terminates when the relative change between successive iterates falls below $\varepsilon_{\text{stop}}$,

$$\frac{\|\mathbf{w}_{k+1} - \mathbf{w}_k\|}{\max(1, \|\mathbf{w}_k\|)} < \varepsilon_{\text{stop}},$$

where the $\max(1, \|\mathbf{w}_k\|)$ normalization makes the test scale-invariant and prevents false triggers when $\|\mathbf{w}_k\|$ is small.

2.5 Convergence analysis

Monotone decrease. As derived in Section 2.2 from the MM framework, IRLS monotonically decreases the objective: $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$ for all k . Since f is bounded below by zero, the sequence $\{f(\mathbf{w}_k)\}$ converges.

Theorem 2.2 (Fixed-point consistency). *If the sequence $\{\mathbf{w}_k\}$ generated by IRLS converges to a point $\mathbf{w}^*$ such that $|(w^*)_i| > \varepsilon_{\text{thr}}$ for all i , then $\mathbf{w}^*$ satisfies the optimality conditions (1.3) of the original problem (2.1).*

Proof sketch. At a fixed point $\mathbf{w}_k = \mathbf{w}_{k+1} = \mathbf{w}^*$, the weighted normal equations (2.9) give:

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + 2\lambda_{\text{IRLS}}\mathbf{W}_*^T\mathbf{W}_*\mathbf{w}^* = 0$$

Since $(\mathbf{W}_*)_{ii} = |w_i^*|^{-1/2}$, we have $(\mathbf{W}_*^T\mathbf{W}_*\mathbf{w}^*)_i = |w_i^*|^{-1}w_i^* = \text{sign}(w_i^*)$, so this reduces to:

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + 2\lambda_{\text{IRLS}} \text{sign}(\mathbf{w}^*) = 0$$

Recalling our mathematically derived relation $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$ from Equation (2.8), we substitute to obtain:

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w}^*) = 0$$

which is exactly the first-order optimality condition $0 \in \partial f(\mathbf{w}^*)$ (equation (1.3)) when all components $w_i^* \neq 0$. $\square$

Convergence rate. In practice, the IRLS scheme used here exhibits linear (geometric) convergence of the iterates: this is consistent with the analysis carried out by Daubechies et al. [4] for the closely related basis-pursuit IRLS algorithm, where local exponential (linear-rate) convergence is established under sparsity and null-space assumptions on the design matrix. The error $\|\mathbf{w}_k - \mathbf{w}^*\|$ thus shrinks by a constant factor at each iteration, set by the condition number of $\mathbf{Q}_k = \mathbf{X}^T\mathbf{X} + 2\lambda_{\text{IRLS}}\mathbf{W}_k^T\mathbf{W}_k$. The extra diagonal contribution $2\lambda_{\text{IRLS}}\mathbf{W}_k^T\mathbf{W}_k \succ 0$ behaves like a Tikhonov-style preconditioner, and typically leaves $\mathbf{Q}_k$ better-conditioned than $\mathbf{X}^T\mathbf{X}$ alone, which is what makes the per-iteration linear solve cheap and reliable.

Verification of hypotheses for the ELM problem. The two assumptions of Theorem 2.2 are satisfied for (2.1) as follows.

1. *Convergence of the sequence.* We work under the assumption that $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}}\mathbf{W}_1^T)$ has full column rank, which is standard for the ELM setting when $M \geq H$ and $\mathbf{W}_1$ is drawn at random with a nonlinear σ [18]. Under this assumption f is strictly convex (Proposition 1.1) and admits a unique minimizer $\mathbf{w}^*$. Since IRLS monotonically decreases f (Section 2.2) and f is bounded below, the sequence $\{f(\mathbf{w}_k)\}$ converges; combined with strict convexity and coercivity of f , standard MM-type arguments for strictly convex surrogates [19] then force the iterates themselves to converge, $\mathbf{w}_k \rightarrow \mathbf{w}^*$.

2. *Non-zero components at convergence.* The hypothesis $|(w^*)_i| > \varepsilon_{\text{thr}}$ holds for the active (non-zero) components of the sparse solution. Components that collapse to zero within ε_{thr} are effectively frozen by the weight clipping (2.5) and drop out of the optimality condition (1.3) — which is exactly the behaviour we want, because these components correspond to features that the L_1 penalty has marked as inactive.

2.6 Computational cost

The per-iteration cost of IRLS breaks down as follows:

- **Weight update $\mathbf{W}_k$:** $O(H)$ — simply iterating over the components of $\mathbf{w}_k$.
- **Matrix update $\mathbf{Q}_k = \mathbf{A} + 2\lambda_{\text{IRLS}}\mathbf{W}_k^T\mathbf{W}_k$:** $O(H)$ — since $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ is pre-computed and only the diagonal changes.
- **Linear system solve $\mathbf{Q}_k\mathbf{w}_{k+1} = \mathbf{b}$:** $O(H^3/3)$ with Cholesky, or $O(pH^2)$ with CG for p CG steps.

The dominant cost per iteration is the linear system solve: $O(H^3)$ with Cholesky. However, the total number of IRLS iterations is typically small (10–50), and the pre-computation of $\mathbf{A} = \mathbf{X}^T\mathbf{X}$ (costing $O(MH^2)$) is amortized over all iterations. The overall cost is therefore:

$$O(MH^2) + k_{\text{IRLS}} \cdot O(H^3)$$

For problems where $H \ll M$ (many training samples, moderate hidden layer size), the pre-computation dominates and IRLS is very efficient.

2.7 Expected behavior on our problem

From the analysis above we expect the following behaviour on the ELM LASSO problem. The objective $f(\mathbf{w}_k)$ decreases at every iteration, with no oscillations, which makes IRLS easy to monitor and to debug: a single non-decreasing step is a red flag. On a semilog plot of $f(\mathbf{w}_k) - f^*$ against k , we expect an approximately straight line, with a slope set jointly by the conditioning of $\mathbf{Q}_k$ and by the safety threshold ε_{thr} .

Sparsity is produced dynamically rather than by an external thresholding step: components $(w_k)_i$ that start out small get assigned increasingly large weights $|(w_k)_i|^{-1}$, which pushes them further toward zero at the next iteration; the stable outcome is a weight vector with many components inside an ε_{thr} -ball of zero. The quantitative trade-offs are then driven by two parameters we chose early on: the regularization strength λ_{LASSO} (the model parameter), and the safety threshold ε_{thr} (a purely numerical one). Since $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$ is fixed by the majorization, the only free knob in the model is λ_{LASSO} : larger values give

sparser solutions but a higher training residual, smaller values do the opposite; conditioning of $\mathbf{Q}_k$ also depends on λ_{LASSO} , so the convergence speed moves with it. For ε_{thr} , smaller values approximate the true L_1 norm more faithfully and yield sharper sparsity, but pay the price in stability whenever a component drifts very close to zero; a value of order 10^{-8} is a reasonable starting point in this trade-off, and we adopt it as our default to be validated experimentally.

Chapter 3

Algorithm 2: Deflected Subgradient Method

In this chapter, we apply the deflected subgradient method with Polyak step and with target level (SGPTL) to the ELM LASSO problem.

3.1 Motivation: why the plain subgradient method is inadequate for ELM LASSO

The ELM LASSO problem

$$f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (3.1)$$

is nonsmooth due to the ℓ_1 penalty. While the plain subgradient method guarantees convergence for convex nonsmooth problems, two structural limitations make it impractical for this problem:

1. **Zig-zagging and slow practical convergence:**

The subgradient update $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{g}_i$ uses only the current subgradient with no memory of previous iterations [16], which on poorly conditioned $\mathbf{X}^T \mathbf{X}$ produces severe oscillations. The complexity lower bound for first-order methods on nonsmooth convex problems is $\Omega(L^2/\varepsilon^2)$ [7, 1], with L a uniform bound on the subgradients. Plain subgradient already attains this optimal rate, but on ELM LASSO the constant L depends on $\|\mathbf{X}\|_2$ and on the radius of the sublevel set containing the iterates, both of which can be large for poorly scaled or ill-conditioned data, so practical convergence is slow.

2. **Stringent stepsize requirements:**

Convergence requires the Decreasing Stepsize Scheduling condition [16], $\sum_i \alpha_i = \infty$ and $\sum_i \alpha_i^2 < \infty$, which forces the steps to shrink quickly (e.g.

$\alpha_i = c/i$) and stalls practical progress on the accuracy targets we care about.

Deflection addresses both issues by injecting memory into the search direction, and is the variant we adopt for ELM LASSO.

3.1.1 The deflection mechanism

Deflected subgradient methods [8] replace the raw subgradient with a direction that blends the current subgradient with the previous search direction:

$$\mathbf{d}_i = \gamma_i \mathbf{g}_i + (1 - \gamma_i) \mathbf{d}_{i-1} \quad (i \geq 1), \quad \mathbf{d}_0 = \mathbf{g}_0, \quad (3.2)$$

where $\gamma_i \in (0, 1]$ is the deflection parameter and the first iteration is handled separately (no previous direction is available). The update becomes $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{d}_i$. Setting $\gamma_i = 1$ at every step recovers the plain subgradient method, whereas $\gamma_i < 1$ gives weight to the memory term $\mathbf{d}_{i-1}$, smoothing out the trajectory. This damps the immediate direction reversals that make the plain subgradient method zig-zag on badly conditioned ELM LASSO instances. We pick γ_i greedily to minimize $\|\mathbf{d}_i\|$, which, coupled with the Polyak-style stepsize introduced below, maximizes the reduction achievable at iteration i [12, 8].

3.2 Convergence framework: balancing deflection and stepsize

For deflected methods on ELM LASSO, the deflection γ_i and stepsize α_i must be carefully coordinated. The theoretical framework is in [8]. We use the stepsize-restricted approach [8]:

$$\alpha_i = \frac{\beta_i (f(\mathbf{w}_i) - f^*)}{\|\mathbf{d}_i\|^2}, \quad \beta_i \leq \gamma_i. \quad (3.3)$$

The condition $\beta_i \leq \gamma_i$ ensures stronger deflection is balanced by smaller steps, preventing large moves in memory-weighted non-descent directions. Since f^* is unknown for ELM LASSO, we replace it with target level $f_{\text{ref}} - \delta$ (Section 3.3), providing an adaptive estimate throughout the run.

Practical handling of $\beta_i \leq \gamma_i$.

We pick a single fixed reference value $\beta \in (0, 1]$ and enforce the condition at each iteration by setting $\beta_i = \min(\beta, \gamma_i)$; the Polyak step in eq. (3.3) uses this clipped β_i . The unrestricted SGPTL pseudocode in [17, Slide 14] admits $\beta \in (0, 2)$, but the stepsize-restricted condition $\beta_i \leq \gamma_i \leq 1$ caps the effective range of the tuning parameter at $(0, 1]$: any $\beta > 1$ would be immediately clipped to γ_i , so allowing larger values brings no benefit. The natural default is $\beta = 1$, which triggers the clipping only when the greedy deflection is already pushing γ_i below one (see Section 3.2.1).

3.2.1 Optimal deflection parameter

At each iteration on ELM LASSO, we choose γ_i to minimize the deflected direction norm:

$$\gamma_i \in \arg \min_{\gamma \in [0,1]} \|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|^2. \quad (3.4)$$

Since the Polyak stepsize has denominator $\|\mathbf{d}_i\|^2$, minimizing this maximizes step length and progress. The problem is quadratic in γ with closed-form solution [8]:

$$\gamma^* = \frac{\|\mathbf{d}_{i-1}\|^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle}{\|\mathbf{g}_i - \mathbf{d}_{i-1}\|^2}, \quad \gamma_i = \min(1, \max(0, \gamma^*)). \quad (3.5)$$

Derivation and interpretation.

The minimization problem in (3.4) is a quadratic function in γ . Expanding $\|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|^2$ yields a parabola in γ ; setting its derivative to zero and solving gives (3.5). The denominator $\|\mathbf{g}_i - \mathbf{d}_{i-1}\|^2$ measures the Euclidean distance between current and previous directions: if they are very different, the denominator is large, making γ^* smaller and emphasizing memory $\mathbf{d}_{i-1}$. The numerator $\|\mathbf{d}_{i-1}\|^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle$ captures alignment: when $\mathbf{g}_i$ and $\mathbf{d}_{i-1}$ point in similar directions, this term is small, again reducing γ^* . Conversely, when $\mathbf{g}_i$ opposes $\mathbf{d}_{i-1}$ (poor alignment), we reduce reliance on the current gradient, dampening oscillations. The clipping $\gamma_i = \min(1, \max(0, \gamma^*))$ ensures $\gamma_i \in [0, 1]$.

3.3 The target-level mechanism

The Polyak stepsize [12] requires knowledge of f^* . The target-level mechanism circumvents this by maintaining an adaptive estimate throughout the run.

The fundamental relationship for subgradient methods [16] is:

$$\|\mathbf{w}_{i+1} - \mathbf{w}^*\|^2 \leq \|\mathbf{w}_i - \mathbf{w}^*\|^2 - 2\alpha_i(f(\mathbf{w}_i) - f^*) + (\alpha_i)^2 \|\mathbf{g}_i\|^2. \quad (3.6)$$

The ideal stepsize $\alpha_i^* = (f(\mathbf{w}_i) - f^*) / \|\mathbf{g}_i\|^2$ [12] maximizes guaranteed reduction but requires f^* .

For ELM LASSO we replace f^* with the target level $f_{\text{ref}} - \delta$ [17], where f_{ref} is the best function value found so far and $\delta > 0$ is the expected improvement; this is the same expression used in the Polyak-step numerator of Algorithm 2. When the target is accurate, the algorithm achieves substantial progress; when it is too optimistic (i.e. $f_{\text{ref}} - \delta < f^*$), the Polyak numerator $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)$ becomes negligible, forcing vanishingly small steps and causing stagnation. We detect stalling by tracking the accumulated displacement $r = \sum \alpha_i \|\mathbf{d}_i\|$: if $r > R$ (the patience threshold) without significant improvement, we contract $\delta \leftarrow \delta \cdot \rho$ with $\rho < 1$ and retry.

3.4 The complete algorithm

Algorithm 2 presents the full Deflected Subgradient Method with target level for ELM LASSO.

Algorithm 2 Deflected Subgradient with Target Level (SGPTL) for ELM LASSO [8]

Require: f (ELM LASSO objective); $i_{\max}$; $\beta \in (0, 1]$; $\delta_0 > 0$; $R > 0$;
 $\rho \in (0, 1)$

- 1: $\mathbf{w}_0 \leftarrow (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ *(same OLS warm start as IRLS)*
- 2: $r \leftarrow 0$; $\delta \leftarrow \delta_0$; $f_{\text{ref}} \leftarrow \bar{f} \leftarrow f(\mathbf{w}_0)$
- 3: **for** $i = 0, 1, \dots, i_{\max}$ **do**
- 4: Compute $\mathbf{g} \in \partial f(\mathbf{w}_i)$ *(subgradient of LASSO)*
- 5: **if** $i = 0$ **then** *(no memory yet)*
- 6: $\gamma_i \leftarrow 1$; $\mathbf{d}_i \leftarrow \mathbf{g}$
- 7: **else**
- 8: Compute γ_i via (3.5) *(optimal deflection)*
- 9: $\mathbf{d}_i \leftarrow \gamma_i \mathbf{g} + (1 - \gamma_i) \mathbf{d}_{i-1}$ *(deflected direction)*
- 10: **end if**
- 11: $\beta_i \leftarrow \min(\beta, \gamma_i)$ *(enforce $\beta_i \leq \gamma_i$)*
- 12: $\alpha_i \leftarrow \beta_i (f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)) / \|\mathbf{d}_i\|^2$ *(Polyak step)*
- 13: $\mathbf{w}_{i+1} \leftarrow \mathbf{w}_i - \alpha_i \mathbf{d}_i$ *(update)*
- 14: $\bar{f} \leftarrow \min(\bar{f}, f(\mathbf{w}_{i+1}))$ *(update record)*
- 15: **if** $f(\mathbf{w}_{i+1}) \leq f_{\text{ref}} - \delta/2$ **then** *(improvement)*
- 16: $f_{\text{ref}} \leftarrow \bar{f}$; $r \leftarrow 0$ *(move checkpoint to current best)*
- 17: **else if** $r > R$ **then** *(stalled)*
- 18: $\delta \leftarrow \delta \rho$; $r \leftarrow 0$
- 19: **else**
- 20: $r \leftarrow r + \alpha_i \|\mathbf{d}_i\|$
- 21: **end if**
- 22: **end for**
- 23: **return** $\mathbf{w}_{\text{best}}$ (iterate achieving $\bar{f}$)

A few notes on the implementation.

Two details of the pseudocode deserve a comment. First, $\mathbf{w}_0 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is the same OLS warm start used by IRLS in Chapter 2; this keeps the two algorithms comparable on the same instance and avoids a cold-start phase in which the Polyak numerator $f(\mathbf{w}_0) - (f_{\text{ref}} - \delta_0)$ could be dominated by a bad guess of δ_0 . Second, the first iteration $i = 0$ is treated separately because (3.5) requires a previous direction $\mathbf{d}_{-1}$ that does not exist yet; setting $\gamma_0 = 1$ matches the initial condition $\mathbf{d}_0 = \mathbf{g}_0$ in eq. (3.2). The naive choice $\mathbf{d}_{-1} = \mathbf{0}$, plugged into (3.5), would yield $\gamma_0 = 0$ and hence $\mathbf{d}_0 = \mathbf{0}$, making the Polyak step ill-defined. The remainder of the loop is standard SGPTL: the record value $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$ is monitored in place of $f(\mathbf{w}_i)$, and the target gap δ is

contracted by $\rho < 1$ whenever no progress is registered over the patience budget R , which removes the need for manual re-tuning of the stepsize when $f_{\text{ref}} - \delta$ drifts below f^* [17].

3.4.1 Parameter calibration for ELM LASSO

SGPTL on ELM LASSO has four parameters: the initial target gap δ_0 , the contraction factor ρ , the patience threshold R , and the step modulation β . The choice of all four will be validated experimentally; here we fix initial ranges and the rationale for them. For δ_0 we take a fraction of the initial objective value, $\delta_0 = c f(\mathbf{w}_0)$ with $c \in [0.01, 0.5]$: small enough that $f_{\text{ref}} - \delta$ does not undershoot f^* at the first steps, large enough to give the algorithm room to travel. For the contraction factor we explore $\rho \in [0.9, 0.95]$, with the upper end of this range giving slower contractions that track f^* more tightly. For the patience threshold we explore R in the range of a few tens of iterations (concretely $R \in [10, 100]$): short enough to react to stalling before wasting a full cycle of non-productive steps, long enough not to trigger on short-horizon plateaus. The step modulation is fixed at $\beta = 1$: the stepsize-restricted variant caps β_i at $\gamma_i \leq 1$ (Section 3.2), so the effective range is $(0, 1]$ and $\beta = 1$ is the most aggressive admissible value. Smaller β remains available as a safety knob in case the runs show instability.

3.5 Application to the ELM LASSO problem

3.5.1 Subgradient computation for ELM LASSO

For the ELM LASSO objective, decompose $f = g + h$ where $g(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2$ (smooth) and $h(\mathbf{w}) = \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$ (nonsmooth).

By the sum rule for subdifferentials [15]:

$$\partial f(\mathbf{w}) = \nabla g(\mathbf{w}) + \lambda_{\text{LASSO}} \partial \|\mathbf{w}\|_1,$$

where $\nabla g(\mathbf{w}) = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$ and the subdifferential of the ℓ_1 norm is component-wise (Section 1.5.3). A subgradient of f at $\mathbf{w}$ is:

$$\mathbf{g} = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \mathbf{s}, \quad s_i \in \partial|w_i|. \quad (3.7)$$

At a component where $w_i \neq 0$ the subdifferential is a singleton and we have no choice but $s_i = \text{sign}(w_i)$; at $w_i = 0$, the subdifferential is the whole interval $[-1, +1]$ and we pick the minimum-norm element $s_i = 0$. This choice is not arbitrary: among all admissible subgradients at $\mathbf{w}$ it is the one with smallest Euclidean norm, which in turn minimizes $\|\mathbf{g}\|^2$ and therefore maximizes the Polyak-step progress $\alpha_i(f(\mathbf{w}_i) - f^*)$ in (3.3).

3.5.2 Convergence analysis and hypothesis verification for ELM LASSO

Theorem 3.1 (Convergence of SGPTL, adapted from [5] and [8]). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. Let $\{\mathbf{w}_i\}$ be the sequence generated by Algorithm 2 with convergence conditions satisfied (stepsize-restricted with $\beta_i \leq \gamma_i$). Assume subgradients are uniformly bounded: $\|\mathbf{g}_i\| \leq L$ for all i . Then:*

1. *The record values $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$ converge: $\bar{f}_i \rightarrow f^*$ as $i \rightarrow \infty$.*
2. *Worst-case complexity is $O(1/\varepsilon^2)$: to achieve $\bar{f}_i - f^* \leq \varepsilon$ requires at most $O(L^2/\varepsilon^2)$ iterations.*

Verification of theorem assumptions for ELM LASSO.

1. **Convexity of $f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$.**

The smooth part g has Hessian $\nabla^2 g = \mathbf{X}^T \mathbf{X} \succeq 0$ (positive semidefinite), so g is convex. The ℓ_1 norm is convex. The sum of convex functions is convex [6], so f is convex.

We assume $\mathbf{X}$ has full column rank, as commonly observed in ELM practice when $M \geq H$ [18]; under this assumption $\nabla^2 g \succ 0$, so g is μ -strongly convex with modulus $\mu = \lambda_{\min}(\mathbf{X}^T \mathbf{X}) > 0$, and adding the convex term h preserves the property [14, 6], so $f = g + h$ is μ -strongly convex as well. In principle this structure could be exploited by methods designed for strongly-convex nonsmooth problems, which improve the worst-case complexity to $O(L^2/(\mu\varepsilon))$ [14]. However, Algorithm 2 does not incorporate any such mechanism: deflection only dampens oscillations, it does not introduce the kind of acceleration or proximal step needed to exploit μ , so the worst-case guarantee for our SGPTL on ELM LASSO remains the standard $O(L^2/\varepsilon^2)$.

2. **Bounded subgradients for ELM LASSO.**

The ℓ_1 component has $\|\lambda_{\text{LASSO}} \mathbf{s}\| \leq \lambda_{\text{LASSO}} \sqrt{H}$ (since each $|s_i| \leq 1$). The smooth component $\mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y})$ grows unboundedly as $\|\mathbf{w}\| \rightarrow \infty$, so f is not globally Lipschitz on $\mathbb{R}^H$.

However, the target-level mechanism prevents arbitrarily large steps: since f is coercive (i.e., $f(\mathbf{w}) \rightarrow \infty$ as $\|\mathbf{w}\| \rightarrow \infty$) and $\bar{f}_i$ is non-increasing, all iterates lie in the compact sublevel set $\{\mathbf{w} : f(\mathbf{w}) \leq f(\mathbf{w}_0)\}$. By Lipschitz continuity on this compact set [10], all subgradients are uniformly bounded by some L depending on the problem data and initial point.

Complexity context: why SGPTL for ELM LASSO.

ELM LASSO is nonsmooth and convex. The intrinsic lower bound is $\Omega(L^2/\varepsilon^2)$ for any first-order method on nonsmooth convex problems [7, 1]. Algorithm 2 achieves this bound, making it theoretically optimal. Deflection does not improve worst-case complexity; its benefit is reducing oscillations and improving practical constants in the O notation.

3.6 Expected practical behavior on the ELM LASSO problem

SGPTL has three features that the experimental chapter will need to take into account. Convergence is *non-monotone*: $f(\mathbf{w}_{i+1}) > f(\mathbf{w}_i)$ occurs frequently and only the record $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$ decreases monotonically to f^* , so semilog plots should display $\log(\bar{f}_i - f^*)$ and the returned iterate is $\mathbf{w}_{\text{best}}$, not $\mathbf{w}_{i_{\text{max}}}$. The rate is sublinear, $O(L^2/\varepsilon^2)$ from Theorem 3.1, so doubling the target precision roughly quadruples the iteration count and accuracies below 10^{-6} are not a realistic goal for this method. The per-iteration cost is $O(MH)$ (two matrix-vector products with $\mathbf{X}$), against the $O(H^3)$ Cholesky solve of IRLS, so DSM is competitive on CPU time only when H is large and the accuracy target is moderate.

Two further points will matter at tuning time. There is no reliable subgradient-based stopping test: unlike the smooth case, where $\|\nabla f(\mathbf{w})\| = 0$ signals optimality, here $\mathbf{w}^*$ satisfies only $0 \in \partial f(\mathbf{w}^*)$ and other elements of the subdifferential at $\mathbf{w}^*$ can have non-zero norm, so $\|\mathbf{g}_i\|$ remaining large is uninformative; we therefore stop on the iteration budget i_{max} or when $\bar{f}_i$ fails to improve appreciably over a fixed window. Sparsity is also different from IRLS: subgradient steps lack the mechanism that pins small components at zero, so DSM produces only approximately sparse iterates, and exact sparsity (when needed) requires a post-termination hard threshold. Finally, the regularization strength λ_{LASSO} enters the subgradient directly through $\mathbf{s}$, so larger values amplify oscillations and slow convergence; the target-level mechanism adapts automatically, but the iteration count is expected to grow with λ_{LASSO} .

Chapter 4

Algorithm Comparison

Both IRLS (Chapter 2) and the Deflected Subgradient Method (Chapter 3) solve the same LASSO problem

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (4.1)$$

yet they differ fundamentally in *how* they handle the non-smoothness of the ℓ_1 penalty and in the trade-offs that result. This chapter compares the two methods along the dimensions that matter most in practice: mathematical strategy, convergence behaviour, computational cost, solution quality, and suitability for the ELM setting.

4.1 Core strategy: how each algorithm handles non-smoothness

The root difficulty in (4.1) is the ℓ_1 term: it is convex but non-differentiable wherever any $w_i = 0$, so classical gradient methods cannot be applied directly. The two algorithms adopt opposite philosophies.

IRLS — smoothing via majorization.

At each iteration IRLS replaces f with a smooth quadratic surrogate $Q(\mathbf{w}, \mathbf{w}_k)$ that majorises f and coincides with it at $\mathbf{w}_k$ (Section 2.2). Minimising the surrogate reduces to a weighted ridge regression problem with a closed-form solution via the normal equations (Section 2.3). The non-smooth ℓ_1 penalty never appears explicitly: it is absorbed into diagonal weights $(\mathbf{W}_k)_{ii}$ that grow large wherever $|w_{k,i}|$ is small, pushing those components toward zero (Section 2.1).

DSM — direct subgradient with memory.

DSM confronts the non-smoothness head-on by descending along a subgradient $\mathbf{g} \in \partial f(\mathbf{w}_i)$ at every step (Section 3.5.1). To suppress the zig-zag oscillations of

the plain subgradient method (Section 3.1), the direction is *deflected*—blended with the previous iterate direction (Section 3.1.1)—and the step length follows a Polyak rule with a self-adapting target-level estimate of f^* (Section 3.3).

In short: IRLS converts the non-smooth problem into a sequence of smooth ones and solves each *exactly* via a linear system, which is what buys it a monotone decrease of f at every iteration. DSM never smooths f ; it leans on the classical subgradient-method convergence theory, pays for it by accepting that $f(\mathbf{w}_i)$ may increase along the way, and keeps track of progress only through the record value $\bar{f}^i$.

4.2 Convergence behaviour

The convergence properties of the two algorithms differ in both *rate* and *monotonicity*.

4.2.1 Monotonicity

The MM framework guarantees that IRLS is **monotone**: $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$ at every iteration (Section 2.2), so a non-decreasing plot of $f(\mathbf{w}_k)$ immediately signals a bug and a simple iterate-change criterion reliably detects convergence. DSM is intrinsically **non-monotone**: individual function values oscillate, and only the record value $\bar{f}^i = \min_{h \leq i} f(\mathbf{w}_h)$ converges monotonically. As a practical consequence, DSM must return $\mathbf{w}_{\text{best}}$ rather than the last iterate, and a fixed iteration budget is the only reliable stopping criterion.

4.2.2 Rate

The two methods occupy different positions in the complexity hierarchy for nonsmooth convex optimisation (Section 3.1). IRLS achieves a **linear (geometric)** rate (Section 2.5): $f(\mathbf{w}_k) - f^*$ decreases by a constant factor each iteration. DSM achieves the information-theoretically optimal **sublinear** rate $O(1/\varepsilon^2)$ for nonsmooth convex functions (Theorem 3.1), so $\bar{f}^i - f^*$ flattens progressively on the same scale. The key quantitative consequence is that improving the target accuracy by one decade requires roughly $100\times$ more DSM iterations, whereas IRLS needs only a constant number of additional steps.

4.2.3 Convergence on the ELM problem

For the ELM LASSO problem $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$ is assumed to have full column rank (the standard ELM condition when $M \geq H$ and $\mathbf{W}_1$ is random with a nonlinear σ), so f is strictly (in fact μ -strongly) convex and admits a unique minimizer $\mathbf{w}^*$. The two algorithms relate to $\mathbf{w}^*$ in different ways: the IRLS iterates $\mathbf{w}_k$ converge to $\mathbf{w}^*$ themselves (Theorem 2.2), at a linear rate governed by the condition number of the weighted normal-equation matrices; for SGPTL, instead, Theorem 3.1 only guarantees that the *record values* satisfy $\bar{f}_i \rightarrow f^*$,

while the iterates $\mathbf{w}_i$ may oscillate by design and need not converge. Strong convexity of f would in principle allow the worst-case complexity to improve from $O(L^2/\varepsilon^2)$ to $O(L^2/(\mu\varepsilon))$ [14], but our SGPTL does not exploit this structure (see Section 3.5.2).

4.3 Computational cost

4.3.1 Per-iteration cost

Operation	IRLS	DSM
Weight / direction update	$O(H)$	$O(H)$
Matrix-vector product(s)	—	$O(MH)$
Linear system solve	$O(H^3)$ (Cholesky) or $O(pH^2)$ (CG)	—
Total per iteration	$O(H^3)$	$O(MH)$

Table 4.1: Per-iteration computational cost of IRLS and DSM.

IRLS requires solving an $H \times H$ symmetric positive definite linear system at every iteration (Section 2.3). With Cholesky factorisation the cost is $O(H^3/3)$ per iteration. With p steps of Conjugate Gradient it is $O(pH^2)$, preferable when H is large and $\mathbf{Q}_k$ is well-conditioned. The matrices $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^T \mathbf{y}$ are pre-computed once at cost $O(MH^2)$ and $O(MH)$ respectively.

DSM requires only two matrix-vector products per iteration (to compute the subgradient: $\mathbf{X}\mathbf{w}_i$ costs $O(MH)$ and $\mathbf{X}^T(\cdot)$ costs $O(MH)$), giving a total per-iteration cost of $O(MH)$ (Section 3.5.1). No precomputation of $\mathbf{X}^T \mathbf{X}$ is needed.

4.3.2 Total cost

Let k_{IRLS} and k_{DSM} denote the respective iteration counts.

$$\begin{aligned} \text{IRLS total: } & O(MH^2) + k_{\text{IRLS}} \cdot O(H^3), \\ \text{DSM total: } & k_{\text{DSM}} \cdot O(MH). \end{aligned}$$

Since IRLS converges linearly and DSM converges sublinearly, the iteration counts satisfy $k_{\text{IRLS}} \ll k_{\text{DSM}}$ for a given accuracy. The regime where DSM is competitive is therefore when H is large enough that $H^3 \gg MH$, i.e. $H^2 \gg M$, so that the Cholesky cost of IRLS dominates.

For the ELM problem, M is the number of training samples and H is the hidden layer size. Typical configurations have $M \gg H$ (many samples, moderate hidden layer), so $O(MH^2) \gg O(H^3)$ and the pre-computation of $\mathbf{A}$ dominates IRLS's total cost. In this regime, the total costs are roughly $O(MH^2)$ for IRLS and $O(k_{\text{DSM}} \cdot MH)$ for DSM, so IRLS is cheaper than DSM whenever

$k_{\text{DSM}} \gtrsim H$. Since Theorem 3.1 forces $k_{\text{DSM}} = O(L^2/\varepsilon^2)$ with L itself growing with H (the ℓ_1 subgradient has norm at most $\lambda_{\text{LASSO}}\sqrt{H}$), the inequality $k_{\text{DSM}} \gtrsim H$ is automatically satisfied at the hidden-layer sizes we consider (any moderate hidden-layer size accessible on a standard machine) for any reasonable accuracy target.

4.4 Solution quality

4.4.1 Sparsity

In the ELM context, sparsity in $\mathbf{w}^*$ identifies which hidden neurons are relevant and which can be pruned. The two algorithms differ sharply in how well they enforce it. IRLS tends to produce numerically sparse solutions as a natural byproduct of its weighted least-squares structure: components near zero accumulate large diagonal weights and are driven to exactly zero (within ε_{thr}) at convergence, without any post-processing (Section 2.7). DSM yields only **approximate sparsity**: subgradient steps carry no mechanism to pin components at exactly zero, so a post-processing thresholding step is typically required (Section 3.6). This distinction is practically significant: IRLS produces interpretable, reproducible sparsity patterns, while DSM’s sparsity depends on the choice of thresholding tolerance.

4.4.2 Accuracy

For a fixed computational budget (number of iterations or wall-clock time), IRLS reaches substantially higher accuracy than DSM on small-to-moderate problems, due to its linear convergence rate. DSM can reach competitive accuracy only when the per-iteration cost advantage of $O(MH)$ versus $O(H^3)$ outweighs the slower convergence rate, i.e. when $H^2 \gg M$ (see Table 4.1).

4.5 Comparison summary for the ELM problem

The two algorithms are therefore complementary rather than interchangeable, and which one we would recommend depends on the regime. On the project’s ELM LASSO problem — fixed random $\mathbf{W}_1$, offline training, M large and H chosen substantially smaller — $M \gg H$ is the default and the $O(MH^2)$ precomputation of $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ is amortized over a small number of cheap $O(H^3)$ Cholesky solves. In this regime IRLS is the natural first choice: it converges linearly, it produces genuinely sparse iterates without a post-processing step, it has a single numerical parameter ε_{thr} to tune, and its monotone $f(\mathbf{w}_k)$ doubles as a cheap correctness check during implementation. DSM becomes competitive when the IRLS precomputation no longer amortizes, i.e. when H grows large enough that $H^2 \gg M$ and the $O(H^3)$ solve dominates, or when memory is tight and we cannot afford to store $\mathbf{X}^T \mathbf{X} \in \mathbb{R}^{H \times H}$: then the $O(MH)$ per-iteration cost and the absence of precomputation tip the balance, provided the accuracy

Criterion	IRLS	DSM
Approach	Smoothing (MM)	Direct subgradient
Monotone	Yes	No (record value only)
Convergence rate	Linear	Sublinear $O(1/\varepsilon^2)$
Per-iteration cost	$O(H^3)$	$O(MH)$
Precomputation	$O(MH^2)$	None
Exact sparsity	Yes	No (needs thresholding)
Parameters to tune	ε_{thr}	δ_0, ρ, R, β
Stopping criterion	Reliable (iterate change)	Unreliable (fixed budget)
Preferred when	High accuracy, $H \ll M$	Large H , moderate accuracy

Table 4.2: Summary comparison of IRLS and DSM on the ELM LASSO problem.

target is moderate (say $\varepsilon \sim 10^{-3}$, beyond which the $O(1/\varepsilon^2)$ rate makes DSM impractical). For the hidden-layer sizes we will actually test (any moderate hidden-layer size accessible on a standard machine), we expect IRLS to win on both accuracy and wall-clock time, and DSM to remain valuable mainly as a baseline and as a sanity check that we are converging to the same $\mathbf{w}^*$.

Chapter 5

Implementation

Chapter 6

Results

Chapter 7

Conclusions

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